

Dr. Leaf

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Analysis Result



Plant/Crop

Identification: Tomato (*Solanum lycopersicum*)

Confidence: High (95%)



Visible Symptoms

- **Widespread Leaf Yellowing (Chlorosis):** Mid-tier and lower leaves show extensive yellowing.
- **Browning & Necrosis:** Leaf edges, tips, and entire lower foliage are brown, dry, and shriveled.
- **Leaf Curling & Wilting:** Foliage appears crispy, curling inward and drooping.
- **Leaf Loss:** Shriveled dead leaves near the lower stem area.
- **Intact Fruit:** Ripening yellow-orange and mature red tomatoes are present and visually

unblemished.



Possible Causes

Possible Cause	Supporting Evidence	Confidence
Natural Senescence / Crop Maturity	Yellowing and leaf drop on lower foliage while fruits are fully formed and ripening; common as tomato plants divert resources to fruit production.	Moderate to High
Nutrient Deficiency (Nitrogen / Potassium)	Nitrogen deficiency causes generalized yellowing starting from older lower leaves. Potassium deficiency causes leaf margins to turn brown and crispy ("scorch").	Moderate
Fungal Disease	Progressive yellowing and	Moderate

Possible Cause	Supporting Evidence	Confidence
(Early Blight / Vascular Wilt)	leaf drying starting from lower leaves upward is characteristic of soil-borne fungal pathogens (<i>Fusarium</i> or <i>Verticillium</i> wilt).	
Water Stress (Under-watering / Irregular Moisture)	Dry, crispy leaf margins and leaf drop can indicate prolonged periods without adequate water.	Moderate



Severity

Moderate

Reason:

While significant foliage yellowing and leaf death are visible, the plant has successfully set fruit that is currently ripening well. The damage primarily affects older leaf tissue rather than the fruit itself. However, if leaf canopy loss becomes severe, it can expose fruit to sunscald or stop further growth.



Recommended Immediate Actions

1. **Harvest Ready Fruit:** Pick mature red and ripening tomatoes promptly to reduce energy burden on the plant and prevent pests or cracking.
 2. **Prune Diseased/Dead Leaves:** Carefully prune off severely yellowed, brown, and dead leaves using sanitized shears. Dispose of them away from the garden (do not compost if fungal disease is suspected).
 3. **Check Soil Moisture:** Check soil moisture 2–3 inches deep. Ensure the plant receives consistent, deep watering at the root zone rather than light, shallow watering.
 4. **Avoid Overhead Irrigation:** Water directly at the base of the plant to keep remaining foliage dry and reduce fungal spore movement.
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Preventive Measures

- **Mulching:** Apply a 2–3 inch layer of organic mulch (straw or wood chips) around the plant base to retain moisture and prevent soil pathogens from splashing onto lower leaves during watering.
- **Crop Rotation:** Rotate nightshade crops (tomatoes, peppers, eggplants, potatoes) to a different location every 3–4 years to avoid build-up of soil-borne wilts.
- **Balanced Nutrition:** Apply a tomato-specific fertilizer balanced in nitrogen, phosphorus,

and potassium, along with micronutrients, throughout the growing season.

- **Proper Spacing & Airflow:** Space tomato plants adequately and trim lower suckers to ensure good air circulation around the canopy.
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When to Consult an Expert

Consult a local agricultural extension service or certified plant pathologist if:

- Entire branches wilt suddenly and collapse while still green.
- Stems show dark brown or black vascular streaks inside when cut.
- The issue spreads rapidly to surrounding healthy plants despite pruning and moisture management.